



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

.NET/WPF PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Desktop

TOTAL: 10 QUESTIONS

Q1 What is .NET/WPF and what problem in Desktop does it solve?

Q6 How does .NET/WPF handle reliability and high availability?

Q2 What are the core components or concepts in .NET/WPF?

Q7 What observability does .NET/WPF provide out of the box?

Q3 How does .NET/WPF compare to its main alternatives?

Q8 What are common performance bottlenecks in .NET/WPF?

Q4 How do you get started with .NET/WPF for a new project?

Q9 What security best practices apply when running .NET/WPF?

Q5 What configuration options most affect .NET/WPF's behavior in production?

Q10 What mistakes do teams commonly make when adopting .NET/WPF?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 .NET/WPF is a tool or platform that

Mitigation

.NET/WPF is a tool or platform that automates or simplifies a specific concern in the Desktop domain, reducing manual effort and operational complexity for engineering teams.

A6 .NET/WPF provides HA through leader election, replication, or stateless horizontal scaling

Availability

.NET/WPF provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 .NET/WPF has key building blocks that work

Primitives

.NET/WPF has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 .NET/WPF exposes metrics (Prometheus endpoint or built-in dashboard)

Observation

.NET/WPF exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 .NET/WPF trades off simplicity against feature richness

Hardening

.NET/WPF trades off simplicity against feature richness differently than its alternatives. Choose .NET/WPF when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install .NET/WPF via its package manager or binary release

Gotchas

Install .NET/WPF via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run .NET/WPF with the principle of least privilege

Assessment

Run .NET/WPF with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels

Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and learning the API by trial and error

Documentation

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.